Universität Bremen

Leibniz-Institut für Präventionsforschung und Epidemiologie – BIPS

Federated Synthetic Data Analysis as a General Approach to Preserve Privacy and Data Structure

A use case using mixed models

Stanley Okafor | Medical Biometry/Biostatistics | Supervisors: Dr. Timm Intemann, Dr. Andreas Mändle

Motivation

Multi-site health research has become increasingly important

- Use of Health data for:
  - data-driven innovation
  - improving public health
  - informing public health policy
- By combining data across research centers, researchers can :
  - increase sample size
  - improve statistical power
  - study rare outcomes
  - improve diversity of study population
  - and obtain findings that are more generalizable

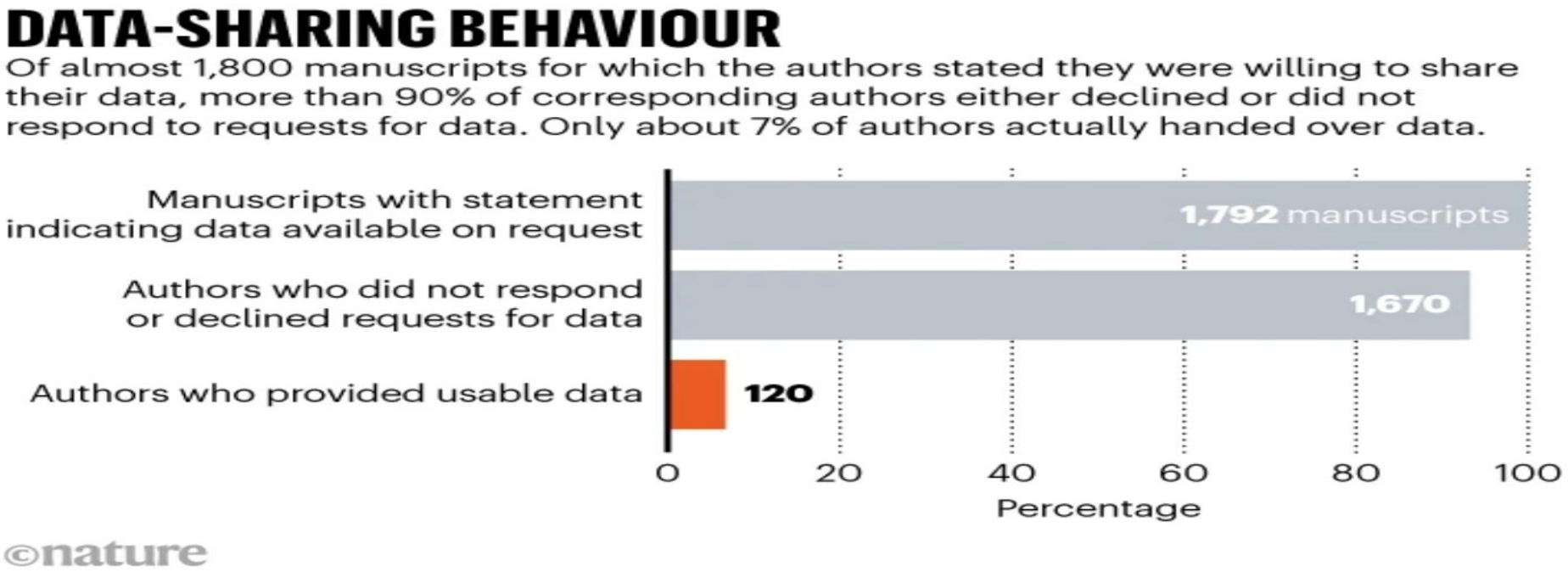


This is particularly relevant in modern health research, where many important research questions cannot be answered adequately using data from single institutions alone

The Problem

Challenges in Data Sharing

- **Privacy Concerns:** Risks of exposing sensitive individual-level data



Source: Watson, Clare. “Many Researchers Say They’ll Share Data — but Don’t.” Nature, vol. 606, no. 7916, June 2022, pp. 853–853. DOI.org (Crossref), https://www.nature.com/articles/d41586-022-01692-1



The World Economic Forum estimates that

97% of hospital data goes unused.

https://www.weforum.org/agenda/2019/12/four-ways-data-is-improving-healthcare

Challenges in Data Sharing

- **Regulations:** Growing number of regulations and policies to protect sensitive data



Europe: General Data Protection Regulation (GDPR)



United States: Health Insurance Portability and Accountability Act (HIPAA)

*Data privacy is driven by this regulators and others, and they exist by continent, country and regions

This restrictions can make centralized pooling of individual-level data difficult or even impossible

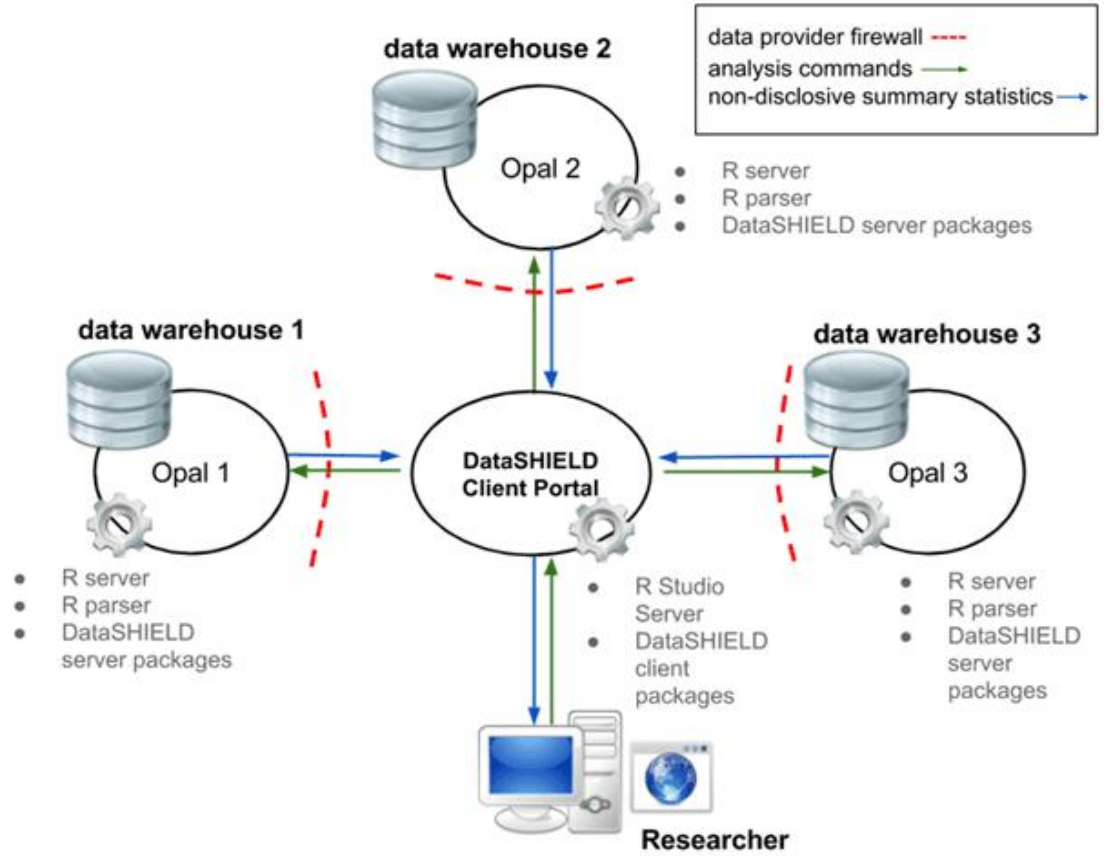
Key Challenge:* *How can we perfom valid multi-site statistical analysis when the original data cannot be shared?

Federated Data Analysis (FDA)

- Individual-level data remain at local sites
- They are never transferred to a central location
- Models are trained collaboratively
- Only aggregate information, such as model estiamtes or summary statistics is shared

- Benefit: Complies with privacy regulations.

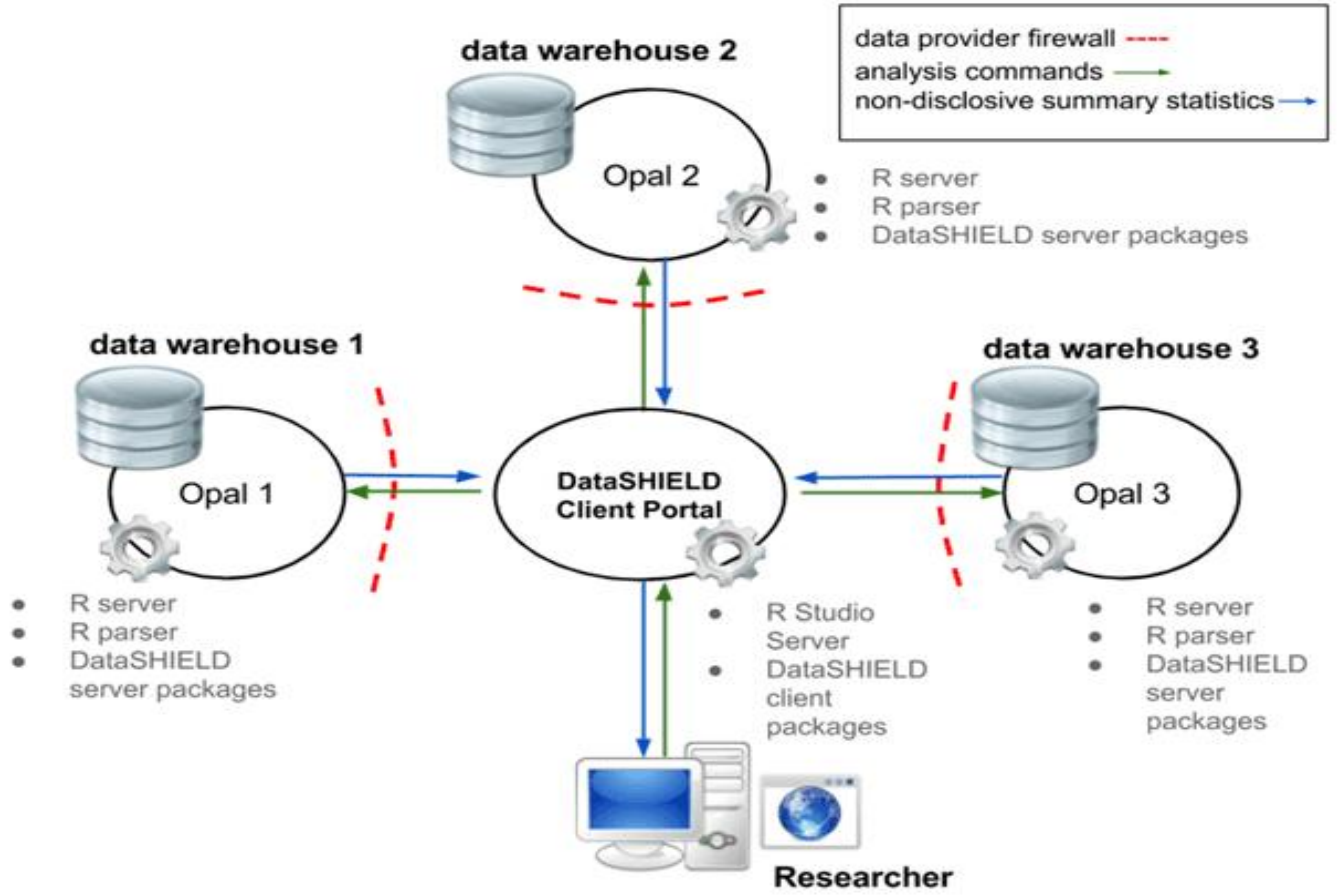
DataSHIELD
Secure Bioscience Collaboration



Source: https://www.datashield.org/about/about-datashield-collated

Challenges of FDA

- Requires specialized infrastructure
- High computational and communication cost between sites
- Time consuming
- Coordination across site is challenging
- Limited flexibility for:
  - complex models



Synthetic Data

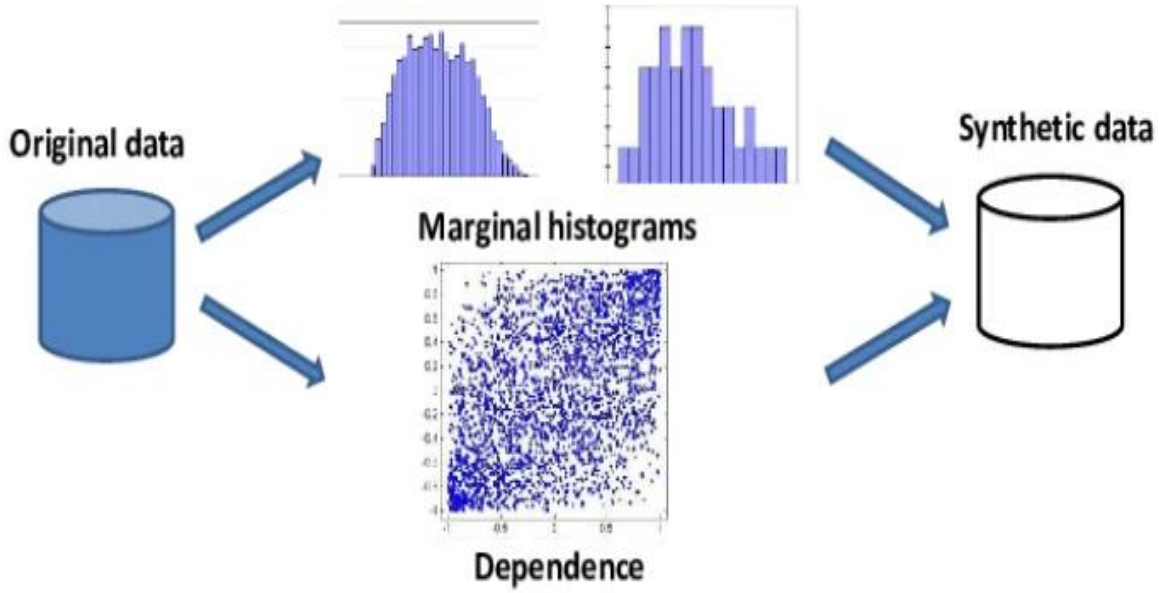
- Artificial or fake data that is generated from the origina data using statistical or machine learning models
- Mimics statistical properties of the original data

Preserves:

- distributions
- correlations
- relationships between variables
- regression patterns

Does not contain:

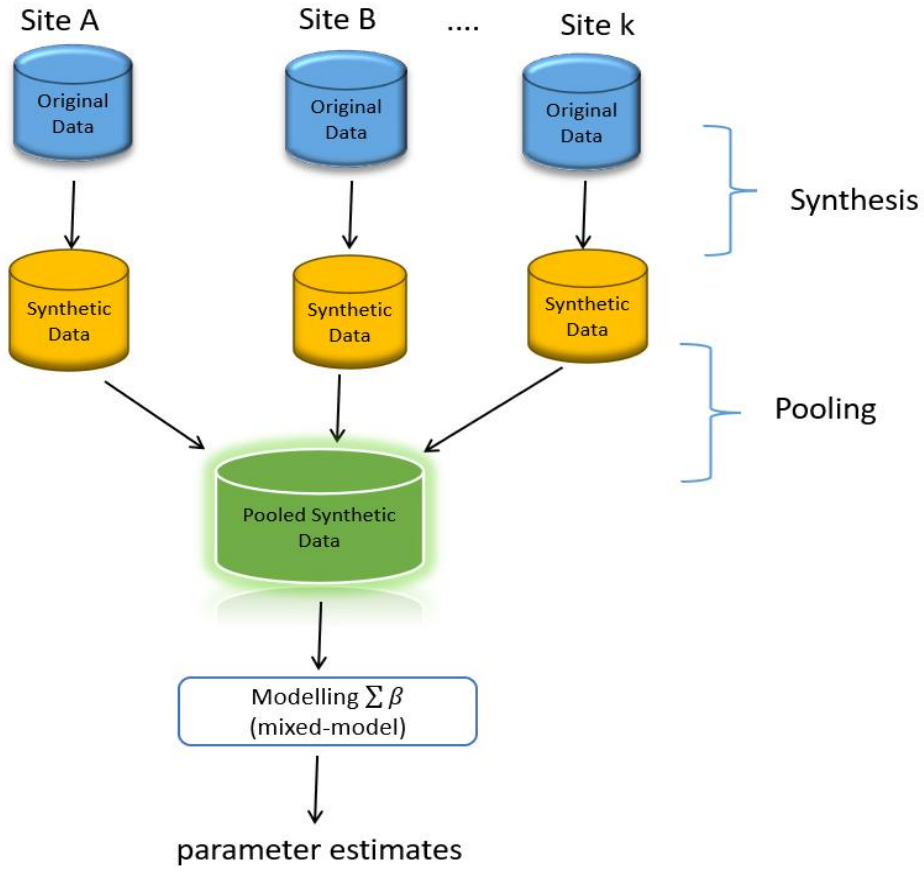
- real individuals information



Source: https://www.researchgate.net/figure/Synthetic-data-generation_fig1_268793256

Federated Synthetic Data Analysis (FSDA)

- Each site generates synthetic data locally
- Synthetic datasets are pooled centrally for analysis



Combines:

- federated setting
- synthetic data

As a result:

- Privacy is preserved
- and researchers can still perform analyses using a centralized synthetic data

Key Research Questions in FSDA

In FSDA key challenges arise:

- Preservation of data structure
  - Correlations and variance components must be retained
- Statistical validity
  - Results should be comparable to the gold standard pooled analysis
- Privacy assurance
  - low re-identification risk (membership disclosure)

Aim & Objectives

Aim:

- Evaluate the performance of FSDA in a multi-site linear mixed effects modelling context

Objectives:

1. Compare FSDA vs pooled analysis (estimates, variance components)
2. Assess robustness (model complexity, heterogeneity)
3. Evaluate membership disclosure risk

Study Design

- To evaluate FSDA we conducted a simulation study
- A simulation framework was chosen because it provides a complete control over the data-generating process and allows the true parameter values to be known in advance

The study consisted of three main stages:

1. Multi-site hierarchical data were generated
2. Synthetic datasets were created independently at each site
3. Statistical analysis were performed on both the original pooled data and the pooled synthetic data and the results were compared

Data Generating Mechanism

Data Simulation

- Multi-site hierarchical structure
- 8 sites (federated setting)
- Varying number of subjects per site (heterogeneity)
- Repeated measurements per subject over time:
  - Baseline (T0)
  - Follow-up (T3, +6 years)
- Nested structure:
  - Observations within subjects
  - Subjects within sites
- Inspired by large cohort studies (IDEFICS / I.Family)

Hierarchical Data Structure

```
Site (highest level)
    ├── Subject 1
    │       ├── Visit 1
    │       ├── Visit 2
    ├── Subject 2
            ├── Visit 1
            ├── Visit 2
```

Covariates (longitudinal)

$$\text{Sex} \sim Bernoulli(0.5)$$

Measured at baseline (T0) and follow-up (T3):

$$\text{Age} : T_0 \sim N(6, 2^2) \rightarrow T_3 = T_0 + 6$$

$$\text{Time Outdoors} : T_0 \sim N(2, 2^2) \rightarrow T_3 \sim N(2, 1.5^2)$$

Random effects (Hierarchical)

$$\text{Site effects} : u_{site} \sim N(0, 3)$$

$$\text{Subject effects} : u_{subject} \sim N(0, 2)$$

Captures:

- ✓ Between-site variation
- ✓ Within-subject correlation

This structure reflects the type of hierarchcical data commonly encountered in health research

Simulation Outcome Model

Equation:

$$Y_{ijk} = \beta_0 + \beta_1 \cdot \text{Sex}_j + \beta_2 \cdot \text{Age}_{ij} + \beta_3 \cdot \text{Time Outdoors}_{ij} + u_{\text{subject}_{j[k]}} + u_{\text{site}_k} + \epsilon$$

Outcome: Y_{ijk} = Vitamin D level (continuous)

$$\text{Deficient}_{ijk} = \begin{cases} 1, & \text{if } Y_{ijk} \leq 50, \\ 0, & \text{if } Y_{ijk} > 50. \end{cases}$$

Model components:

- Fixed effects: Sex, Age, Time Outdoors
- Random effects: $\mathbf{u}_{subject}, \mathbf{u}_{site}$
- Residual error: ϵ

True parameters (used for evaluation)

$$\beta_0 = 45,\ \beta_1 = 2,\ \beta_2 = -0.5,\ \beta_3 = 1.5$$

$$\sigma^2_{\text{site}} = 3^2,\ \sigma^2_{\text{subject}} = 2^2,\ \sigma^2_{\epsilon} = 14^2$$

This prespecified values to induce realistic associations between the outcome, covariates and variance components to capture hierarchical structure

Synthetic Data Generation (SDG)

Sequential Conditional Modeling Framework (synthpop)

Key Idea:

Instead of modeling the full joint distribution directly:

$$P(X_1, X_2, \ldots, X_p)$$

the sequential conditional modeling framework is essentially decomposing the joint distribution into a series of conditional distributions:

$$P(X_1, \ldots, X_p) = \prod_{j=1}^{p} P(X_j \mid X_1, \ldots, X_{j-1})$$

- Full joint distribution approximated using sequential conditional models
- Variables synthesized one at a time
- Each variable conditioned on previous synthesized variables
- Preserves multivariate dependence structure

Application in FSDA:

for illustration purpose:

P(Site_ID)
× P(u_site | Site_ID)
× P(nSubjects | Site_ID, u_site)
× P(Subject_ID | Site_ID, nSubjects)
× P(u_subject | Site_ID, Subject_ID)
× P(Sex | previous variables)
× P(Visit | previous variables)
× P(Age | previous variables)
× P(Time Outdoors | previous variables)
× P(Vitamin D Level | previous variables)
× P(Deficient | Vitamin D)

This ordering ensures that downstream variables inherit the dependencies established earlier in the synthesis process

Site_ID: Site Identification, Subject_ID: Subject Identification, u_site: site random effect, u_subject: subject random effect, nSubjects: number of subjects

Key Design Choices in Synthesis

Synthesis Strategy 06

Whether proper synthesis or simple synthesis should be incorporated?

- whether uncertainty is incorporated into

Synthesizing Model 05

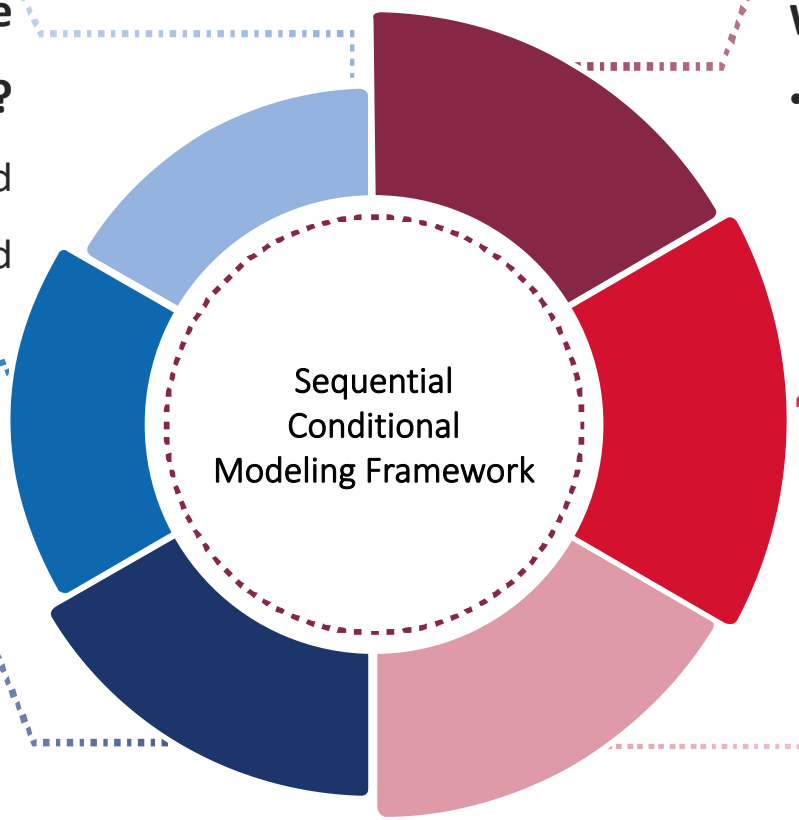
Which synthesis model is used?

- Which synthesizer estimates the conditional distribution

Nested Synthesis 04

Whether nesting is preserved?

- Preservation of hierarchical structure



01 Order of Variables

Which variables comes first?

- which conditional models is estimated first

02 Predictor Matrix

Which predictors are used?

- which previous variables enter each conditional model

03 Stratified Synthesis

Whether synthesis is stratified?

- whether synthesis occurs separately by groups

Key Design Choices in Synthesis

01

Variable Ordering

Determines the synthesis sequence

$$X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow \cdots$$

and how dependencies are propagated

$$\text{Identifiers} \rightarrow \text{Covariates} \rightarrow \text{Outcome}$$

So later variables condition on previously synthesized variables.

02

Predictor Matrix

The framework theoretically allows:

$$P\left(X_j \mid X_1, \ldots, X_{j-1}\right)$$

But in practice:

$$P\left(X_j \mid X_2, X_5, X_7\right)$$

not all previous variables must be used

The predictor matrix specifies:

- which predictors actually enter each conditional model
- and which variables are excluded

controls model complexity, dependency preservation, and overfitting.

Key Design Choices in Synthesis

03

Stratified Synthesis

Instead of estimating

$$P(X_j \mid \cdot)$$

globaly,

Synthesis was performed separately within each site

$$P(X_j \mid \cdot, \text{Site} = s)$$

This preserves:

- site-specific distributions
- between-site heterogeneity

04

Nested Synthesis

Instead of:

$$P(\text{Subject ID})$$

we preserved hierarchical structure by conditioning lower-level identifiers on higher-level groups

$$P(\text{Subject ID} \mid \text{Site ID})$$

This preserves

- clustering
- multilevel dependence structure

Key Design Choices in Synthesis

05

Synthesizer Model

Sequential conditional model only specifies

$$P(X_j \mid \cdot)$$

But does not specify how to estimate it.

Here we used:

- CART - primarily here for continuous variables
- Logistic regression - binary variables

06

Synthesis Strategy (Proper Synthesis)

Instead of fixing model parameters at a single estimate

$$\hat{\theta}$$

Proper synthesis approximates the posterior predictive distribution

instead of computing the integral directly:

$$p(Y_{\text{syn}} \mid Y_{\text{obs}}, X_{\text{obs}}) = \int p(Y_{\text{syn}} \mid X_{\text{obs}}, \theta)\, p(\theta \mid Y_{\text{obs}}, X_{\text{obs}})\, d\theta$$

using Monte Carlo sampling we repeatedly draw parameters from their posterior distribution $\theta^{(m)} \sim p(\theta \mid Y_{\text{obs}}, X_{\text{obs}})$

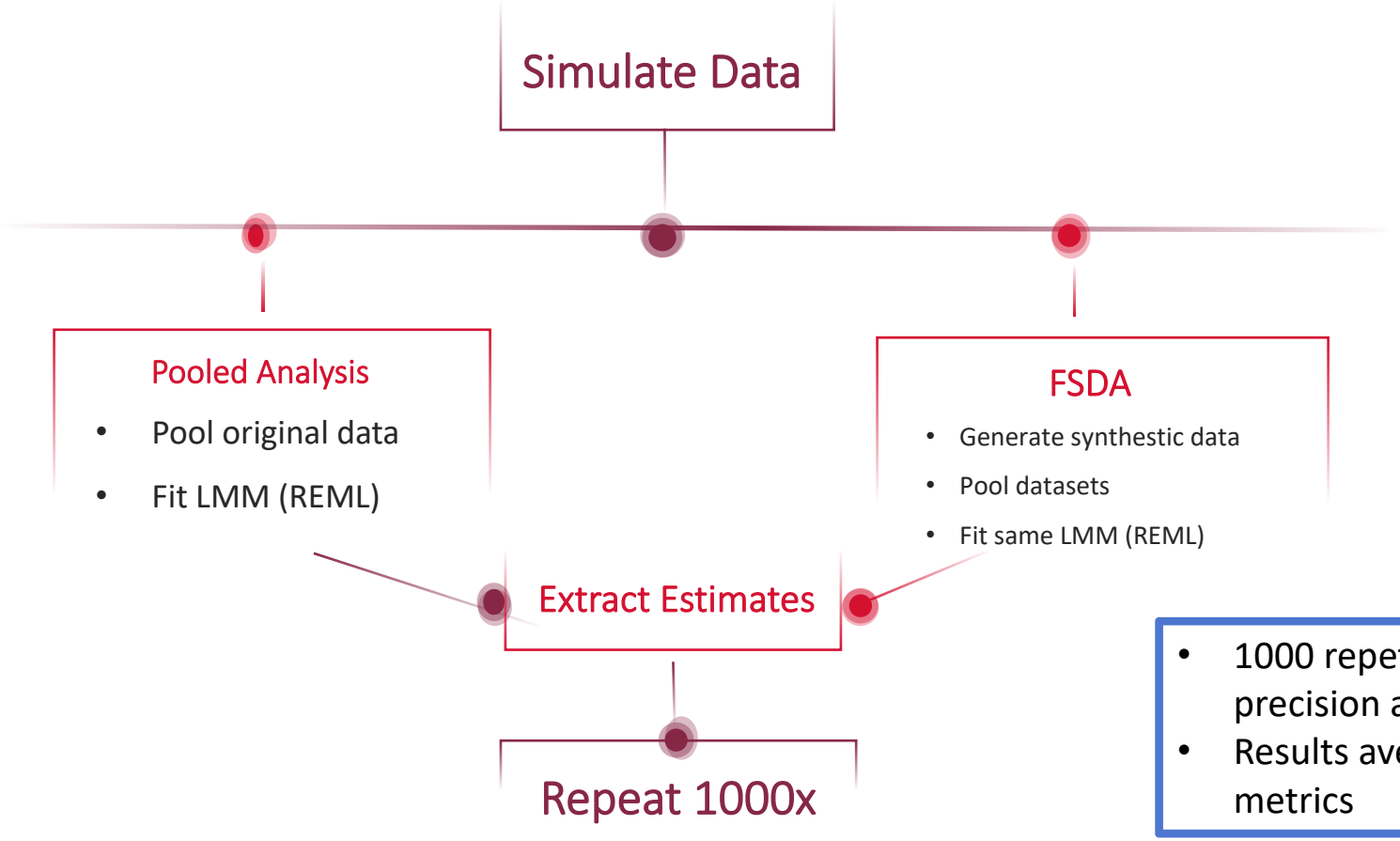
we generate synthetic data conditional on that draw

$$Y_{\text{syn}}^{(m)} \sim f(Y_{\text{syn}} \mid X_{\text{obs}}, \theta^{(m)})$$

So each synthetic dataset is generated using its own sampled parameter set,

thereby propagating uncertainty and variablity of the released synthetic data.

Simulation Based Comparison Framework



- 1000 repetitons wa choosen to balance precision and computational feasibility
- Results averaged to compute performance metrics

LMM: Linear Mixed Model, REML: Restricted Maximum Likelihood, FSDA: Federatd Synthetic Data Analysis

Model Fitting & Implementation

- Models: Linear Mixed Models (LMM) was fitted to both data
- Estimation: The model was fitted using Restricted maximum likelihood (REML) estimation
- Software:
  - R

01

simstudy

data generation

02

synthpop

- synthetic data generation
- Utility assessment
- membership disclosure risk

03

lme4

model fitting

- 1000 simulation repetitions
- parallel compution

Performance Measures

Bias

$$\text{Bias}(\hat{\theta}) = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i - \theta$$

Measures accuracy (closeness to true value)

Relative Bias

$$\text{Rel. Bias}(\hat{\theta}) = \frac{\text{Bias}(\hat{\theta})}{\theta} \times 100\%$$

Standardizes the bias relative to the true parameter allowing for comparison across parameters

Variance

$$\text{Var}(\hat{\theta}) = \frac{1}{n_{\text{sim}} - 1} \sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \bar{\hat{\theta}})^2$$

Measures stability across simulations (precision)

MSE

$$\text{MSE}(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$$

Combines accuray + precision (overall balance of performance)

computed across 1000 simulation replicates for both pooled analysis and FSDA

Measuring Utility: Propensity Score (pMSE)

Original data

Sex	Age	Time Outdoors	Vitamin D level
0	6	2.1	45
1	8	1.5	52

Synthetic data

Sex	Age	Time Outdoors	Vitamin D level
0	6.2	2.0	44
1	7.9	1.6	50

Combined data

Sex	Age	Time Outdoors	Vitamin D level	I	$\hat{p}_r$
0	6	2.1	45	0	0.48
1	8	1.5	52	0	0.45
0	6.2	2.0	44	1	0.52
1	7.9	1.6	50	1	0.55

Classifier (model) sees:
Predictors = variables
Target = 1 (0 = original, 1 = synthetic)
It learns this patterns

if data are different:
it predicts:
~ 0 for original
~ 1 for synthetic
if data are very similar:
it predicts probabilities near 0.5

$$\text{pMSE} = \frac{1}{N}\sum_{r=1}^{N}\left(\hat{p}_r - c\right)^2$$

The pMSE measures how far these predicted probabilities are from the true proportion

$\hat{p}_r$: model's predicted probability that row r is synthetic

c : true proportion of synthetic data $\left(c = \frac{n_{\text{syn}}}{N} \approx 0.5\right)$

So a small pMSE means the classifier cannot distinguish the datasets, indicating high utility

Standardized pMSE

Raw pMSE depends on:

- Sample Size (N)
- Class balance (c)
- Model complexity (k)

➢ pMSE values not directly comparable across settings

$$\mathbb{E}_0[\text{pMSE}] = \frac{(k-1)(1-c)^2 c}{N}$$

$$\text{SD}_0[\text{pMSE}] = \frac{\sqrt{2(k-1)}(1-c)^2 c}{N}$$

with equal proportion c = 0.5, this simplifies to:

$$\mathbb{E}_0[\text{pMSE}] = \frac{(k-1)}{8N}, \quad \text{SD}_0[\text{pMSE}] = \frac{\sqrt{2(k-1)}}{8N}$$

$$S_\text{pMSE} = \frac{\text{pMSE} - \mathbb{E}_0[\text{pMSE}]}{\text{SD}_0[\text{pMSE}]}$$

Measures deviation from:

What we expect under correct synthesis, allowing interpretation relative to a null benchmark

- S_pMSE ≈ 0 - good synthesis
- Large values - poor similarity

Overall S_pMSE not computed (CART - no closed-form null distribution)
Variable-level SpMSE computed using logistic models

Disclosure Risk Assessment

Identity disclosure risk- RepU (Replicated Uniques)

Can an intruder identify individuals from synthetic data?

$$RepU = 100 \times \frac{\sum_{q \in Q} (s_{.q} \mid d_{.q} = 1 \ \wedge \ s_{.q} = 1)}{N_d}$$

% of individuals who are:

- Unique in original data
- AND also unique in synthetic data

High - High re-indentification risk ✈

Low - Good privacy protection ☼

Quasi-identifiers (q):
(Site_ID, Sex, Age, Visit, Time Outdoors)

Count how many people fall into that group:
In original data - *d.q*
In synthetic data - *s.q*

Continues variables (Age, Time Outdoors, Vitamin D level)

- Grouped into bins to avoid artificial uniqueness
  - UiO ≈ 100% ✈
  - RepU ≈ 0 ✈

q	Count in Original	Count in Synthetic
A, M, 10-12, T0, Low(0-2)	1	1
A, F, 10-12, T0, Low(0-2)	3	2

First row is unique (d.q = 1 ∧ s.q = 1)

Disclosure Risk Assessment

Attribute disclosure risk - DiSCO

Can an intruder infer sensitive information from synthetic data?

$$DiSCO = 100 \times \frac{\sum_{q \in Q} \sum_{t} (d_{tq} \mid p_{stq} = 1)}{N_d}$$

% of cases where:

- Synthetic data predicts a sensitive value
- AND prediction is correct in original data

High - Sensitive info can be inferred ✈

Low - Safe ☼

Target variable: Vitamin D

Group (q):

q	Vitamin D level
A, M, 10-12, T0, Low(0-2)	45
A, M, 10-12, T0, Low(0-2)	45

Synthetic always says:

q	Vitamin D level
A, M, 10-12, T0, Low(0-2)	always 45

- All have same value
- Intruder can correctly infer Vitamin D

Results: Simulated Data Characteristics

- n ≈ 80,000 subjects
- 8 sites
- 2 time points (T0, T3)
- Balanced Sex (~50%)
- Age: 6 → 12 years
- Time Outdoors: stable (~2 hrs/day)

- Outcome: Vitamin D
- Mean ≈ 44-47 nmol/L

Table 3.1: *Descriptive characteristics of the simulated dataset (n ≈ 80,000 subjects across 8 sites). Continuous variables are summarized with means, standard deviations (SD), and ranges; categorical variables are summarized with proportions.*

Continuous variables	Mean	SD	Min	Max
Age_{T0}	5.99	2.00	-2.38	14.1
Age_{T3}	12.00	2.00	3.62	20.1
Time $Outdoors_{T0}$	1.99	2.00	-6.67	10.8
Time $Outdoors_{T3}$	2.01	1.49	-4.75	8.23
Vitamin D $level_{T0}$	47.0	14.5	-11.4	104.
Vitamin D $level_{T3}$	44.1	14.4	-16.6	105.
Categorical variables				
Sex (Male)		50.1%		
Deficient ($Y \leq 50$)		60.4%		

Age, Time Outdoors, and Vitamin D level were simulated from normal distributions, which resulted in small proportion of negative values (Age: 0.16% at T0; 0% at T3; Time Outdoors: 16.0% at T0; 8.9% at T3; Vitamin D level: 0.06% at T0; 0.10% at T3), which were retained for analysis.

Results: Simulated Data Characteristics

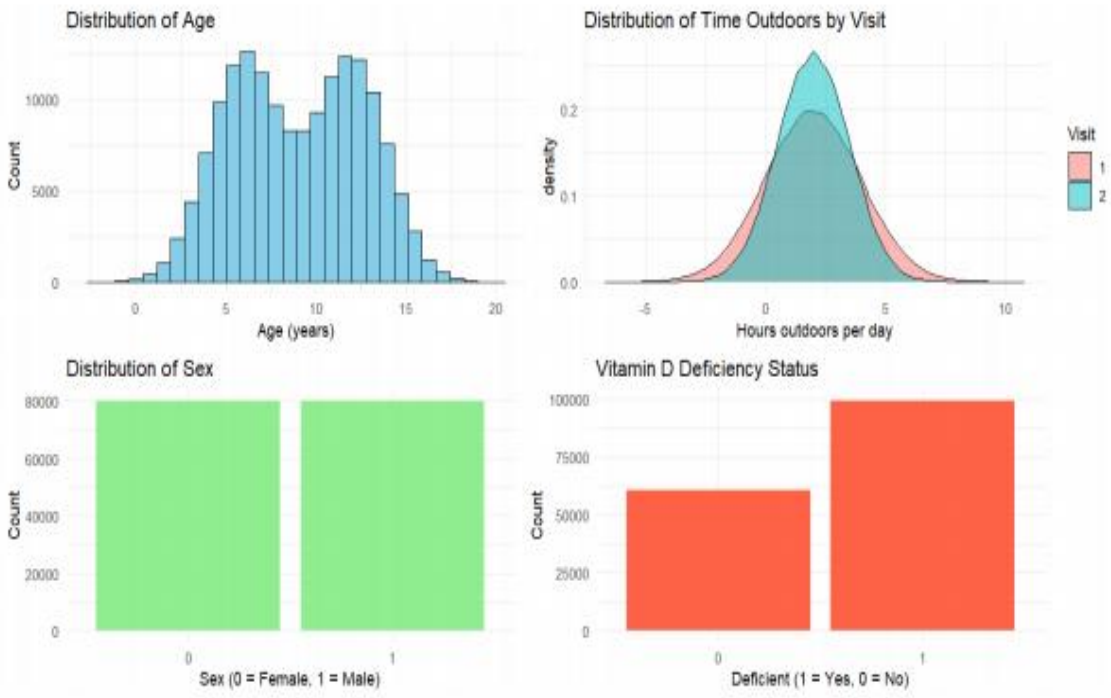


Figure 3.2: *Distributions of baseline covariates in the simulated dataset. (A) Age across all time points, showing bimodal distribution due to measurement at baseline (T0) and follow-up (T3). (B) Daily hours spent outdoors at baseline (T0) and follow-up (T3). (C) Sex distribution (0 = female, 1 = male). (D) Vitamin D deficiiency status, defined as $Y \leq 50$ nmol/L.*

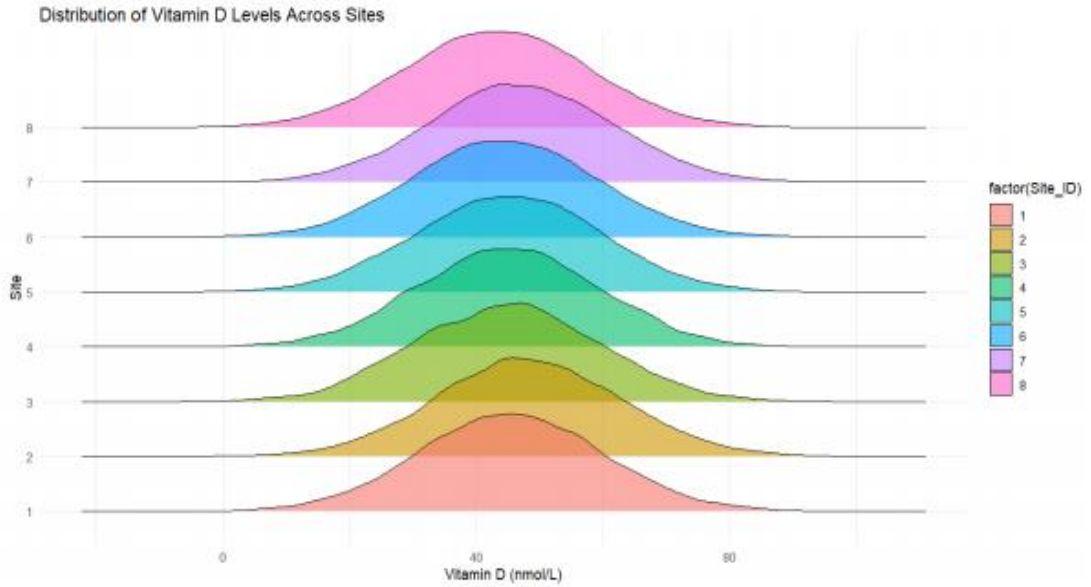


Figure 3.1: *Distribution of Vitamin D levels across the 8 simulated sites. Each ridge represents the density for one site, illustrating site-specific shifts in the outcome distribution. Although the distributions appear similar overall, the clustering reflects the imposed site-level random intercept variance ($\sigma^2_{site} = 3^2$), which induces variability in Vitamin D levels between sites.*

These properties confirm that the simulation reflects a realistic multi-site longitudinal setting

Synthetic Data Preserves Distributions

Original Data:

Table 3.1: *Descriptive characteristics of the simulated dataset ($n \approx 80{,}000$ subjects across 8 sites). Continuous variables are summarized with means, standard deviations (SD), and ranges; categorical variables are summarized with proportions.*

Continuous variables	Mean	SD	Min	Max
Age_{T0}	5.99	2.00	-2.38	14.1
Age_{T3}	12.00	2.00	3.62	20.1
Time $Outdoors_{T0}$	1.99	2.00	-6.67	10.8
Time $Outdoors_{T3}$	2.01	1.49	-4.75	8.23
Vitamin D $level_{T0}$	47.0	14.5	-11.4	104.
Vitamin D $level_{T3}$	44.1	14.4	-16.6	105.
Categorical variables				
Sex (Male)		50.1%		
Deficient ($Y \leq 50$)		60.4%		

Age, Time Outdoors, and Vitamin D level were simulated from normal distributions, which resulted in small proportion of negative values (Age: 0.16% at T0; 0% at T3; Time Outdoors: 16.0% at T0; 8.9% at T3; Vitamin D level: 0.06% at T0; 0.10% at T3), which were retained for analysis.

Synthetic Data:

Table 3.2: *Descriptive characteristics of the synthesized dataset ($n \approx 80{,}000$ subjects across 8 sites). Continuous variables are summarized with means, standard deviations (SD), and ranges; categorical variables are summarized with proportions.*

Continuous variables	Mean	SD	Min	Max
Age_{T0}	6.01	1.99	-2.12	13.5
Age_{T3}	12.0	2.00	3.62	19.9
Time $Outdoors_{T0}$	2.01	1.93	-6.67	10.4
Time $Outdoors_{T3}$	2.01	1.57	-5.79	9.47
Vitamin D $level_{T0}$	46.9	14.5	-11.6	102.
Vitamin D $level_{T3}$	44.0	14.4	-11.6	105.
Categorical variables				
Sex (Male)		50.5%		
Deficient ($Y \leq 50$)		62.2%		

Age, Time Outdoors, and Vitamin D level were synthesized based on the original distributions, which led to a small proportion of negative values (Age: 0.16% at T0; 0% at T3; Time Outdoors: 14.7% at T0; 9.6% at T3; Vitamin D level: 0.08% at T0; 0.11% at T3). These values were retained in the synthesized dataset to preserve the overall statistical properties of the original data.

- Marginal distributions well reproduced
- Mean & SD matched
- Sex and deficiency proportions similar

Results: Utility

- Synthetic data ≠ identical data
- Goal: distributional similarity

Overall pMSE = 0.217 (low)

- Most variables:
- pMSE ≈ 0
- S_pMSE ≈ 0

→ High utility

- Classifier cannot distinguish datasets
- High similarity in joint distribution

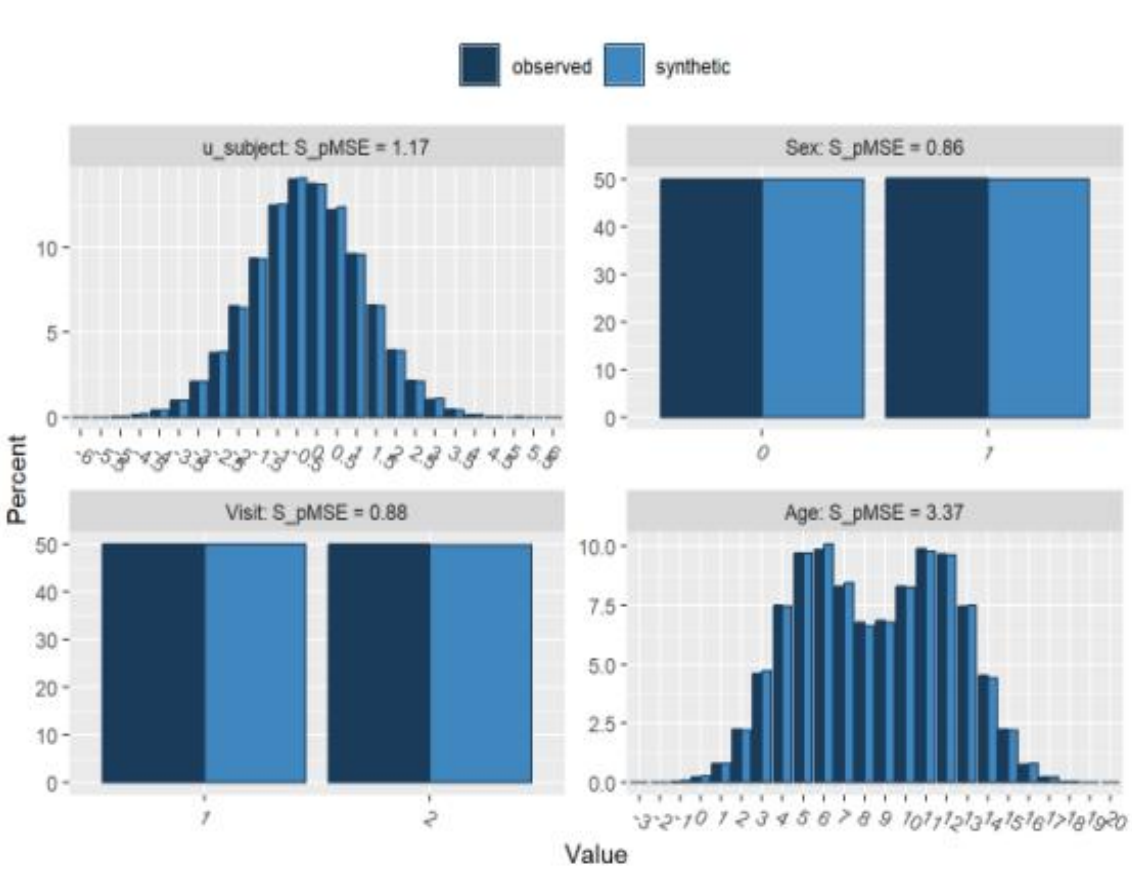
Table 3.3: *Variable-specific and overall utility measures comparing synthetic and original datasets, calculated using CART propensity score models. Low pMSE and S_pMSE indicate high similarity between synthetic and original data.*

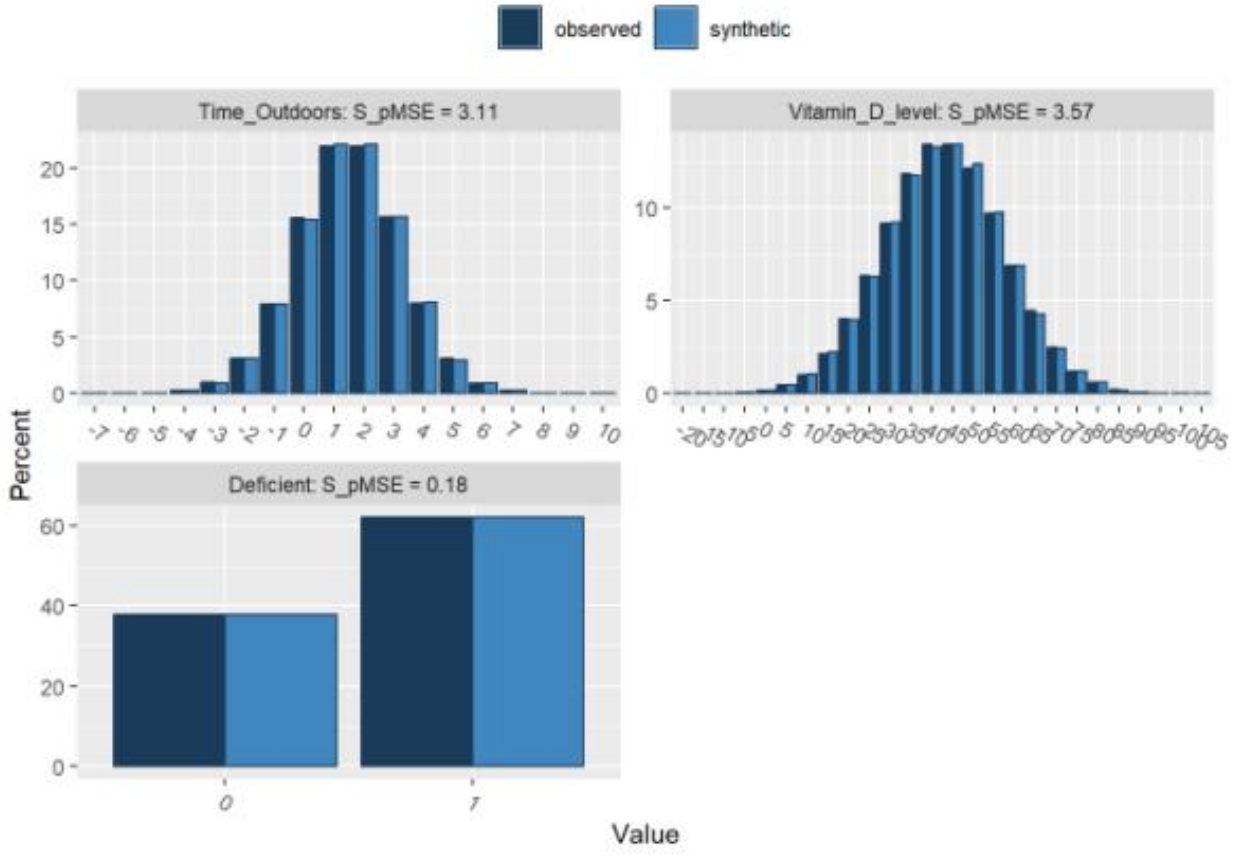
Variable	pMSE	S_pMSE	df
Site_ID	0.000000	0.201173	3
u_site	0.058939	37720.98	4
nSubjects	0.000000	0.280008	3
Subject_ID	0.000007	4.287210	4
u_subject	0.000002	1.171947	4
Sex	0.000000	0.855625	1
Visit	0.000000	0.883601	1
Age	0.000005	3.366109	4
Time_Outdoors	0.000005	3.106851	4
Vitamin_D_level	0.000006	3.572126	4
Deficient	0.000000	0.182979	1
Overall (utility.gen)	0.216659	NA	–

Note: The inflated S_pMSE for `u_site` reflects the small number of site categories (8) and is not indicative of poor utility. Overall, pMSE values indicate that the synthetic data closely match the original data. df = degree of freedom.

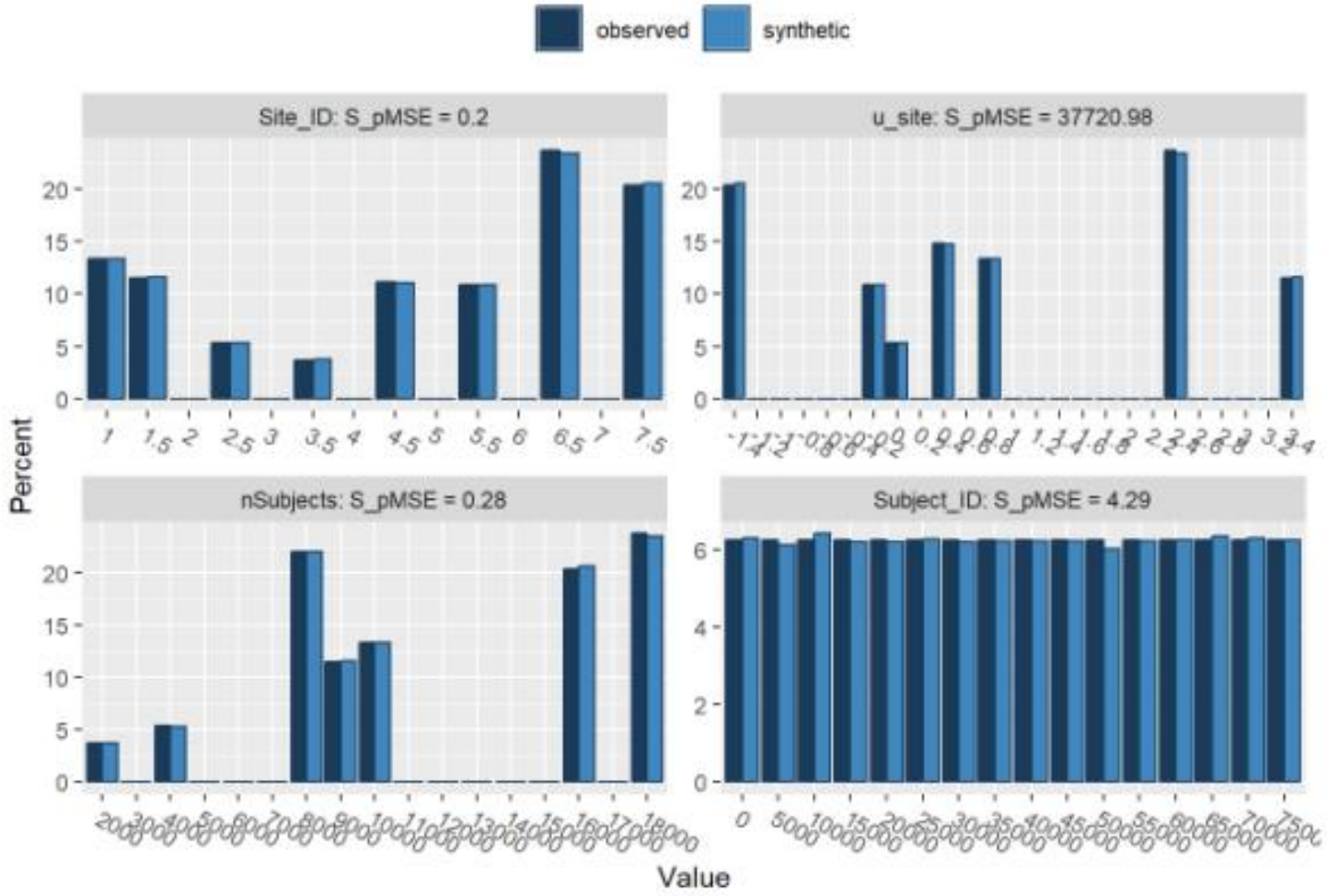
u_site: high S_pMSE due to small categories (not poor utility)

Results: Utility





Results: Utility



Results show:

- ✓ Relationships preserved
- ✓ Hierarchical structure maintained
- ✓ Strong similarity between original and synthetic data

Results: Disclosure Risk (Privacy Protection)

Goal: Assess whether synthetic data can reveal identities or sensitive attributes (information)

Key findings:
RepU: 0 - 0.01%
DiSCO (Subject_ID): 0%
DiSCO (Vitamin D): 0 - 0.05%
DiSCO (Deficient): 0 - 0.20%

Results show:
- ✓ Very low identity disclosure ≈ 0
- ✓ Almost no original individuals can be re-identified
- ✓ Very low attribute disclosure
- ✓ Strong privacy protection across all sites

Recommended disclosure measures (Raab et al., 2024)

Site	RepU (%)	DiSCO: Subject_ID (\%)	DiSCO: Vitamin D (%)	DiSCO: Deficient (%)
Site 1	0.01	0.00	0.00	0.13
Site 2	0.01	0.00	0.00	0.03
Site 3	0.00	0.00	0.03	0.18
Site 4	0.00	0.00	0.05	0.20
Site 5	0.01	0.00	0.00	0.00
Site 6	0.01	0.00	0.00	0.00
Site 7	0.01	0.00	0.01	0.04
Site 8	0.00	0.00	0.01	0.00

Key finding: RepU ≈ 0% and DiSCO ≈ 0% ⇒ Minimal identity and attribute disclosure risk

Results: FSDA vs Gold Standard Pooled Analysis

Table 3.8: *Side-by-side comparison of mean estimates, bias, relative bias, variance, and MSE between the gold standard pooled analysis and FSDA across all parameters.*

(a) *Mean estimate and Bias (part a)*

Parameter	True Value	Mean (Pooled)	Mean (FSDA)	Bias (Pooled)	Bias (FSDA)
Fixed Effects					
Intercept	45.0	45.084	46.097	0.084	1.097
Sex	2.0	1.927	1.528	-0.073	-0.472
Age	-0.5	-0.488	-0.484	0.012	0.016
Time Outdoors	1.5	1.479	1.468	-0.021	-0.032
Random Effects					
Subject-level variance	2.0	1.503	1.984	-0.497	-0.017
Site-level variance	3.0	2.980	2.374	-0.020	-0.630
Residual					
Residual variance	196.0	196.641	196.162	0.641	0.162

Fixed Effects

- Both methods recovered:
  - Age
  - Time Outdoors
  - and Intercept
    with minimal bias.
- The pooled analysis showed smaller bias overall, while FSDA showed greater underestimation for:
  - Sex coefficient

Random Effects

- The pooled analysis underestimated the subject-level variance
- FSDA recovered the subject-level variance more closely to the true value
- In contrast, FSDA underestimated the site-level variance

Residual Variance

Both methods recoverd the residual variance accurately, with only small deviation from the true value

Results: FSDA vs Gold Standard Pooled Analysis

(b) *Relative Bias, Variance and MSE (part b)*

Parameter	Rel. Bias (%) (Pooled)	Rel. Bias (%) (FSDA)	Var (Pooled)	Var (FSDA)	MSE (Pooled)	MSE (FSDA)
Fixed Effects						
Intercept	0.19	2.4	0.374	0.034	0.381	1.236
Sex	-3.66	-23.6	0.000	0.002	0.005	0.238
Age	-2.39	-3.2	0.000	0.000	0.000	0.001
Time Outdoors	-1.40	-2.1	0.000	0.001	0.000	0.002
Random Effects						
Subject-level variance	-24.86	-0.8	0.000	0.194	0.247	0.194
Site-level variance	-0.68	-20.9	2.415	0.035	2.415	0.427
Residual						
Residual variance	0.33	0.1	0.000	1.256	0.411	1.282

Relative Bias

- FSDA showed larger relative bias for:
  - Sex
  - Site-level variance.
- However,
  pooled analysis also showed notable rel. bias for:
  - Subject-level variance

Variance Across 1000 Repetitions

- poold analysis showed high variability in site-level variance estimation
- FSDA showed lower variability for site-level variance, but more systematic underestimation.

MSE

- Fixed effects generally showed low MSE for both methods
- Differences for both methods were mainly observed for variance components.

Results: Key Observations

Utility

- Synthetic data closely reproduced the distributions and multivariate structure of the original data.
- Low pMSE values suggested high statistical similarity between datasets.

Disclosure Risk

- RepU and DiSCO values remained very low across sites and target variables.
- This indicated minimal observed identity and attribute disclosure risk.

Inferential Performance

- FSDA recovered most fixed effects and residual variance reasonably well when compared to the pooled analysis.
- Differences were mainly observed for:
  - the Sex coefficient
  - and variance components.

Discussion: Interpretation of Key Findings

Fixed Effects

- Intercept, Age, and Time Outdoors recovered with relatively small bias
- Consistent with literature showing synthetic data can preserve strong associations
- Suggests FSDA retained key inferential relationships

Subject-Level Variance

- Recovered closer to true value by FSDA
- Likely related to CART partitioning and Bayesian bootstrap sampling
- Similar robustness of tree-based methods has been reported in previous studies

Site-Level Variance

- Challenging for both methods
- Only 8 sites available
- Consistent with established small-cluster variance estimation literature

Synthesizer Effects

- Bias in Sex coefficient likely related to logistic synthesis model
- Indicates FSDA performance depends on synthesizer choice
- Supports recommendations for careful synthesizer selection

Overall, the observed differences were largely attributable to known statistical and synthesis-related challenges that are consistent with existing literature rather than fundamental limitations of FSDA

Why Was the Sex Effect More Biased?

- Sex was synthesized using logistic regression rather than CART
- Different synthesizers preserve relationships differently
- Logistics models may perform less well when data exhibits complex dependencies

Interpretation:

- The larger bias observed for the Sex coefficient likely reflects the synthesis model selected for that variable
- Suggests FSDA performance depends not only on the framework itself but also on synthesizer choice (Hayes et al., 2015; Nowok et al., 2016, Reiter et al. 2003, Reiter et al. 2005b)

Differences in inference may arise from synthesizer specification rather than from FSDA itself

Why Was Subject-Level Variance Estimation Challenging Under Low SNR?

- Residual variance was intentionally large
- Makes separation of:
  - subject-level variance
  - residual variance

  statistically difficult
- Both methods were affected by the limited signal available for variance decomposition

Interpretation:

- When noise dominates the outcome , variance components become difficult to estimate precisely
- Small changes in the data can be allocated either to random effects or to residual error
- This can lead to biased or unstable variance estimates, even when the model is correctly specified
- Consistent with established mixed-model literature (e.g., Pinheiro & Bates; Snijders & Bosker)

Some observed differences likely reflect the challenging data conditions rather than shortcomings of FSDA itself

Why Was Site-Level Variance Difficult to Recover?

- Estimation of cluster-level variance is known to be unstable when the number of clusters is small
- Limited information is available for accurately separating between-site variability from samplying variation

Observed pattern:

- Pooled analysis produced site-variance estimates close to the true value on average, but with substantial variability across repetitions
- FSDA produced more stable estimates, but tended to underestimate between-site variance
- Lower MSE resulted from a bias-variance trade-off

Interpretation:

- Variance component estimation becomes less reliable when cluster counts are small
- Both findings are consistent with literature on small-cluster mixed models and synthetic data generation (McNeish & Stapleton, 2016; Drechsler & Reiter, 2011)

Limited cluster information likely contributed to the observed differences in site-level variance estimation for both methods

Why Did FSDA Recover Subject-Level Variance Closer to the True Value?

- CART synthesizes data by recursively partitioning the predictor space
- Synthetic values are generated within terminal nodes using Bayesian bootstrap samplying
- This process may preserve local subject-level heterogeneity even when residual noise is high
- In contrast, variance decomposition in mixed effects models becomes difficult when residual variance dominates the signal

Interpretation:

- FSDA estimates were centered much closer to the true subject-level variance
- Although variability increased across repetitions, overall MSE remained lower than pooled analysis
- Similar robustness of tree-based synthesizers under noisy conditions has been discussed in the synthetic data literature

The CART-based synthesis may help preserve subject-level heterogeneity under noisy conditions, allowing FSDA to recovr subject-level variance closer to the true value and acheive lower overall estimation error than pooled analysis

e

Why Did Some Differences Occur?

Overall:

Differences between FSDA and pooled analysis largely driven by:

- low SNR
- small cluster count
- and synthesizer model choice

rather than complete inferential failure of FSDA.

Limitations

01

Low Signal-to-Noise Ratio

- Large residual variance relative to subject-level variance
- Difficult separation of subject and residual variability

May reduce generalizability to higher-SNR settings.

02

Small Number of Sites

- Only 8 sites included
- Site-level variance estimation maybe unstable

Performance may differ with larger number of clusters.

03

Synthesizer Choice

- Different synthesizers used across variable types.
- Logistic synthesis for Sex introduced bias

FSDA sensitive to synthesizer specification

04

Simplified Simulation Design

- Random intercept model only
- No random slopes, interactions or complex longitudinal structure
- Assumed exchangeable site effects

Real-world data may exhibit greater site heterogeneity and non-exchangeable structures.

Results should therefore be interpreted within the specific simulation and modeling conditions investigated in this thesis

Practical Implications

Main Points

FSDA is promising for privacy-preserving multicenter analysis

- Useful when direct data sharing is restricted
- Preserves:
  - population-level (fixed-effect) inference
  - maintains multilevel regression structure

Important Practical Insight

Variance components are sensitive under:

- low SNR
- few clusters
- complex hierarchical dependence

Therefore:

FSDA is most reliable when inference focuses on

- population-level effects (fixed effects)
- overall associations across sites
- marginal inference

Future Work and Recommendations

Recommended Areas

- Evaluate larger numbers of sites/clusters
- Compare alternative synthesizers:
  - CART
  - Random Forest
  - Bayesian Networks
  - GANs/VAEs
- Explore Bayesian hierarchical models
- Extend FSDA to:
  - Survival models
  - generalized mixed models
  - nonlinear models
- Apply FSDA to real clinical datasets

Conclusions

Key Findings

- ✓ FSDA preserved the multilevel structure and key statistical relationships in the original data
- ✓ Synthetic data achieved high utility and low disclosure risk
- ✓ Inferential results were broadly comparable to the gold standard pooled analysis

Variance Components

- ✓ Variance component estimation remained challenging under the low-SNR, small-cluster setting
- ✓ When bias and variability were considered jointly (MSE), FSDA achieved lower overall error for both subject-level and site-level variance components

Under the simulation conditions investigated, FSDA achieved a practical balance between:

- Analytical Utility
- Privacy Protection
- and Inferential Validity

and represents a promising framwork for privacy-preserving multi site health research where centralized data shearing is restricted

References

Ahrens, W., Siani, A., Adan, R., De Henauw, S., Eiben, G., Gwozdz, W., Hebestreit, A., Hunsberger, M., Kaprio, J., Krogh, V., Lissner, L., Molnár, D., Moreno, L.A., Page, A., Picó, C., Reisch, L., Smith, R.M., Tornaritis, M., Veidebaum, T., Williams, G., Pohlabeln, H., Pigeot, I., on behalf of the I.Family consortium: Cohort Profile: The transition from childhood to adolescence in European children–how I.Family extends the IDEFICS cohort. International Journal of Epidemiology. dyw317 (2016). https://doi.org/10.1093/ije/dyw317

Azizi, Z., Lindner, S., Shiba, Y., Raparelli, V., Norris, C.M., Kublickiene, K., Herrero, M.T., Kautzky-Willer, A., Klimek, P., Gisinger, T., Pilote, L., El Emam, K.: A comparison of synthetic data generation and federated analysis for enabling international evaluations of cardiovascular health. Scientific Reports. v. 13, p. 11540 (2023). https://doi.org/10.1038/s41598-023-38457-3

Bates, D., Mächler, M., Bolker, B., Walker, S.: Fitting Linear Mixed-Effects Models Using lme4. Journal of Statistical Software. 67, (2015). https://doi.org/10.18637/jss.v067.i01

Breiman, L., Friedman, J.H., Olshen, R.A., Stone, C.J.: Classification And Regression Trees. Routledge (2017)

References

Caiola, G., Reiter, J.P.: Random forests for generating partially synthetic, categorical data. Transactions on Data Privacy. 3, 27–42 (2010)
Drechsler, J.: Synthetic Datasets for Statistical Disclosure Control: Theory and Implementation. Springer New York, New York, NY (2011)
Drechsler, J., Reiter, J.P.: An empirical evaluation of easily implemented, nonparametric methods for gener_x0002_ating synthetic datasets. Computational Statistics & Data Analysis. 55, 3232–3243 (2011). https://doi.org/10.1016/j.csda.2011.06.006
El Emam, K., Mosquera, L., Fang, X.: Validating a membership disclosure metric for synthetic health data. JAMIA Open. v. 5, p. ooac083 (2022). https://doi.org/10.1093/jamiaopen/ooac083
Hastie, T., Tibshirani, R., Friedman, J.: The Elements of Statistical Learning. Springer New York, New York, NY (2009)
Hayes, T., Usami, S., Jacobucci, R., McArdle, J.J.: Using Classification and Regression Trees (CART) and random forests to analyze attrition: Results from two simulations. Psychology and Aging. 30, 911–929 (2015). https://doi.org/10.1037/pag0000046
Laird, N.M., Ware, J.H.: Random-Effects Models for Longitudinal Data. Biometrics. 38, 963 (1982). https://doi.org/10.2307/2529876

References

Little, R.J.A.: Statistical analysis of masked data. Journal of Official Statistics. 9, 407–426 (1993)
McNeish, D., Stapleton, L.M.: Modeling Clustered Data with Very Few Clusters. Multivariate Behavioral Research. 51, 495–518 (2016). https://doi.org/10.1080/00273171.2016.1167008
Morris, T.P., White, I.R., Crowther, M.J.: Using simulation studies to evaluate statistical methods. Statistics in Medicine. 38, 2074–2102 (2019). https://doi.org/10.1002/sim.8086
Nowok, B., Raab, G.M., Dibben, C.: Synthpop: Bespoke creation of synthetic data in r. Journal of Statistical Software. v. 74, pp. 1–26 (2016). https://doi.org/10.18637/jss.v074.i11
Pinheiro, J.C., Bates, D.M.: Mixed-effects models in S and S-PLUS. Springer, New York Berlin Heidelberg (2004)a
Pinheiro, J.C., Bates, D.M.: Mixed-effects models in S and S-PLUS. Springer, New York Berlin Heidelberg (2004)b
Raab, G.M., Nowok, B., Dibben, C.: Guidelines for Producing Useful Synthetic Data, http://arxiv.org/abs/1712.04078, (2017)
Raab, G.M., Nowok, B., Dibben, C.: Assessing, visualizing and improving the utility of synthetic data, https://arxiv.org/abs/2109.12717, (2021)
Raab, G.M., Nowok, B., Dibben, C.: Practical privacy metrics for synthetic data, https://arxiv.org/abs/2406.16826, (2024)
Raghunathan, T.E., Reiter, J.P., Rubin, D.B.: Multiple imputation for statistical disclosure limitation. Journal of Official Statistics. 19, 1–16 (2003). https://doi.org/10.2478/v10010-007-0001-5

References

Reiter, J.P.: Inference for partially synthetic, public use microdata sets. Survey Methodology. 29, 181–188 (2003)
Reiter, J.P.: Releasing Multiply Imputed, Synthetic Public use Microdata: An Illustration and Empirical Study. Journal of the Royal Statistical Society Series A: Statistics in Society. 168, 185–205 (2005)a. https://doi.org/10.1111/j.1467-985X.2004.00343.x
Reiter, J.P.: Using CART to generate partially synthetic, public use microdata. Journal of Official Statistics. 21, 441–462 (2005)b
Reiter, J.P.: Using CART to generate partially synthetic, public use microdata. Journal of Official Statistics. 21, 441–462 (2005)c
Rubin, D.B.: The Bayesian Bootstrap. The Annals of Statistics. 9, (1981). https://doi.org/10.1214/aos/1176345338
Rubin, D.B.: Multiple Imputation for Nonresponse in Surveys. Wiley (1987)
Rubin, R.B.A.: Discussion: Statistical disclosure limitation. Journal of Official Statistics. 9, 461–468 (1993)
Snijders, T.A.B., Bosker, R.J.: Multilevel analysis: An introduction to basic and advanced multilevel modeling. SAGE, Los Angeles, Calif. (2012)
Snoke, J., Raab, G.M., Nowok, B., Dibben, C., Slavkovic, A.: General and Specific Utility Measures for Synthetic Data. Journal of the Royal Statistical Society Series A: Statistics in Society. 181, 663–688 (2018). https://doi.org/10.1111/rssa.12358

References

Studnicki, M., Piepho, H.P.: Hierarchical modelling of variance components makes analysis of resolvable incomplete block designs more efficient. Theoretical and Applied Genetics. 137, 134 (2024). https://doi.org/10.1007/s00122-024-04639-4

Verbeke, G.: Linear Mixed Models for Longitudinal Data. Springer New York, New York, NY (2001)

Verbeke, G., Molenberghs, G.: Linear mixed models for longitudinal data. Springer, New York, NY (2011)

Watson, D.S., Blesch, K., Kapar, J., Wright, M.N.: Adversarial random forests for density estimation and generative modeling. In: Ruiz, F., Dy, J., and Meent, J.-W. van de (eds.) Proceedings of the 26th international conference on artificial intelligence and statistics. pp. 5357–5375. PMLR (2023)

Wilson, R.C., Butters, O.W., Avraam, D., Baker, J., Tedds, J.A., Turner, A., Murtagh, M., Burton, P.R.: DataSHIELD – New Directions and Dimensions. Data Science Journal. 16, 21 (2017). https://doi.org/10.5334/dsj-2017-021

Wolters, M., Intemann, T., Russo, P., Moreno, L.A., Molnár, D., Veidebaum, T., Tornaritis, M., De Henauw, S., Eiben, G., Ahrens, W., Floegel, A.: 25-Hydroxyvitamin D reference percentiles and the role of their determinants among European children and adolescents. European Journal of Clinical Nutrition. 76, 564–573 (2022). https://doi.org/10.1038/s41430-021-00985-4

Yan, Z., Zachrison, K.S., Schwamm, L.H., Estrada, J.J., Duan, R.: A privacy-preserving and computation_x0002_efficient federated algorithm for generalized linear mixed models to analyze correlated electronic health records data. PLOS ONE. v. 18, p. e0280192 (2023). https://doi.org/10.1371/journal.pone.0280192

Thank You
Questions & Discussion